

Topic 5: The Deterministic Rules Engine — KDIGO AKI & Drug Matrices

TOPIC OBJECTIVE

In life-critical clinical decision support, non-deterministic generative models cannot be trusted with diagnostic thresholding. In src/rules.py, AegisClinical implements mathematically verifiable clinical guidelines: formal KDIGO 2012 Acute Kidney Injury staging, physiological lab panic thresholds, and an $O(1)$ set-intersection pharmacological interaction matrix that catches lethal contraindications.

1. The Real-World Analogies: Bridge Strain & The Garden Hose

A. The Bridge Strain Gauge (KDIGO AKI)

Imagine a suspension bridge engineered for a baseline load of 100 tons (baseline creatinine = 1.0 mg/dL). If heavy traffic increases the load to 150 tons (+50%, 1.5x ratio), the strain gauges sound an initial warning (**Stage 1 AKI**). If the load surges to 240 tons (2.4x ratio), the support cables begin to tear (**Stage 2 AKI**). At 300+ tons, the structure collapses (**Stage 3 AKI**). Engineers don't prompt a generative chatbot to guess the bridge's safety—they read hard strain sensors.

B. The Garden Hose Pressure Collapse (The Triple Whammy)

Imagine a garden hose with water flowing through to water a lawn (the kidney glomerulus):

- **1. The Diuretic** turns down the faucet at the house, reducing overall water volume in the hose.
- **2. The NSAID (Ibuprofen)** steps firmly on the hose at the entrance (afferent arteriole constriction via prostaglandin blockade).
- **3. The ACE-Inhibitor (Lisinopril)** opens the nozzle wide at the exit (efferent arteriole dilation via Angiotensin II blockade).

The Result: Filtration pressure inside the kidney drops to zero. Glomerular filtration ceases, and the patient goes into acute renal failure.

2. Mathematical KDIGO 2012 AKI Staging Algorithm

In src/rules.py, evaluate_kdigo_aki(baseline_cr, current_cr) implements the internationally accepted KDIGO guidelines:

KDIGO Stage	Mathematical Criteria	Severity	Mandatory Clinical Action
Stage 1	Ratio $\geq 1.5x$ (within 7d) OR Delta $\geq +0.3$ mg/dL (within 48h)	MODERATE	Review volume status, monitor fluids, discontinue nephrotoxins.
Stage 2	Ratio 2.0x to 2.9x baseline creatinine	HIGH	Urgent renal function review; stop all ACEI/ARBs, NSAIDs, and diuretics.

Stage 3	Ratio >= 3.0x baseline OR Current Cr >= 4.0 mg/dL with acute rise >= 0.5	CRITICAL	Immediate emergent nephrology consultation; prepare for potential dialysis.
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3. The O(1) Drug Interaction Matrix

Rather than doing quadratic $O(N^2)$ pairwise comparisons, AegisClinical normalizes medication strings and checks set intersections against pre-compiled pharmaceutical classes in **O(1) amortized time**:

```
# Normalize strings: 'Lisinopril 20 MG Oral Tablet' -> 'lisinopril' norm =
normalize_drug_name(raw_name) # Set intersection evaluates lethal combinations in O(1)
matched_ace = normalized_set.intersection(ACE_ARBS) matched_diuretic =
normalized_set.intersection(DIURETICS_LOOP_THIAZIDE) matched_nsaid =
normalized_set.intersection(NSAIDS) if matched_ace and matched_diuretic and matched_nsaid: #
Trigger 'Triple Whammy' CONTRAINDICATED alert
```

4. Concrete Patient Case Walkthroughs

Case 1: Arthur Morales (Patient 01)

- Baseline Creatinine: 1.0 mg/dL | Current Creatinine: 2.4 mg/dL
- Math: Ratio = 2.4 / 1.0 = 2.4x -> **KDIGO Stage 2 AKI (HIGH)**
- Medications: Lisinopril (ACE) + Furosemide (Diuretic) + Ibuprofen (NSAID) -> **Triple Whammy (CONTRAINDICATED)**

Case 2: Elena Rostova (Patient 02)

- Serum Potassium: 5.2 mEq/L (Normal high: 5.0) -> **Hyperkalemia Alert (HIGH)**
- Medications: Losartan (ARB) + Spironolactone (Potassium-sparing Diuretic) -> **Dual Aldosterone Cascade Blockade (MAJOR)**
- Directive: Discontinue spironolactone or adjust dose; order urgent ECG to inspect for cardiac arrhythmias.

KNOWLEDGE CHECK & NEXT STEP

- 1. Why is the 'Triple Whammy' classified as CONTRAINDICATED rather than MODERATE?**
Because the simultaneous collapse of renal blood flow causes acute tubular necrosis requiring emergency intervention.
- 2. What is the computational complexity of the drug lookup matrix?**
By hashing drug names into Python sets, intersection operations execute in O(1) amortized time, scaling to any formulary size.
- 3. Next Step in Topic 6:** LangGraph & State Contracts — Parallel Reducers with operator.add.